

ShopFlow AWS Project

SNS Design

Pub/Sub Topics - Fan-Out Pattern - Ops Alerts

Module 13 of 18

Field	Value
Version	1.0
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Author	Manish Kumar Sharma
Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon SNS provides publish/subscribe messaging for ShopFlow. One message published to a topic is delivered to all subscribers simultaneously (fan-out). ShopFlow uses SNS for two purposes: broadcasting order events to multiple SQS queues, and delivering CloudWatch alarm notifications to the ops team.

2. Topics

Topic Name	Subscribers	Publisher	Purpose
shopflow-ops-alerts-dev	manishkumarsharma24@gmail.com	CloudWatch Alarms	Alert ops team when alarm fires
shopflow-order-fanout-dev	shopflow-notifications SQS + shopflow-order-events SQS	Spring Boot	Fan-out single order event to multiple consumers

3. SNS Fan-Out Pattern

Step	Component	Action
1	Spring Boot	Publishes order event to shopflow-order-fanout-dev topic
2	SNS to Notification SQS	Copy delivered to shopflow-notifications-dev
3	SNS to Order Events SQS	Copy delivered to shopflow-order-events-dev
4	Lambda (email)	Reads from notifications SQS, sends email
5	Lambda (order processor)	Reads from order events SQS, updates inventory

4. SQS Queue Policy for SNS

SNS must be granted permission to send messages to each SQS queue via a Queue Policy. ShopFlow uses a Condition to restrict the source to the specific SNS topic ARN:

```
Effect: Allow | Principal: sns.amazonaws.com
Action: sqs:SendMessage
Condition: aws:SourceArn = arn:aws:sns::...:shopflow-order-fanout-dev
```

5. SNS vs SQS

Attribute	SNS	SQS
Pattern	Pub/Sub -- one to many	Queue -- one to one
Persistence	No -- fire and forget	Yes -- messages wait to be consumed
Retry	No (at-least-once attempt)	Yes -- visibility timeout + retries
Best combo	SNS + SQS	Gets both fan-out AND durability

6. Email Subscription -- Action Required

The ops-alerts topic has an email subscription for manishkumarsharma24@gmail.com. After CloudFormation deploys, AWS sends a confirmation email. The confirmation link MUST be clicked before any alerts are delivered.

Action Required	When
Click 'Confirm subscription' in the email from no-reply@sns.amazonaws.com	Within 72 hours of stack deployment -- link expires